

ISM Note 5

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1. Radiative process (continued)

Bremsstrahlung radiation (free-free radiation)

The Bremsstrahlung radiation is the radiation emitted by the free electrons when they are accelerated by the Coulomb field of the ions. The power emitted by a single electron is given by

$$P = \frac{2e^2|a|^2}{3c^3}$$

where a is the acceleration of the electron.

The emission of a large group of free electrons follows the Maxwell-Boltzmann distribution. Totally the emission follows

$$\epsilon_\nu \propto Z^2 n_e n_i T^{-1/2} e^{-\frac{h\nu}{kT}}$$

And the power spectrum plot is as follows

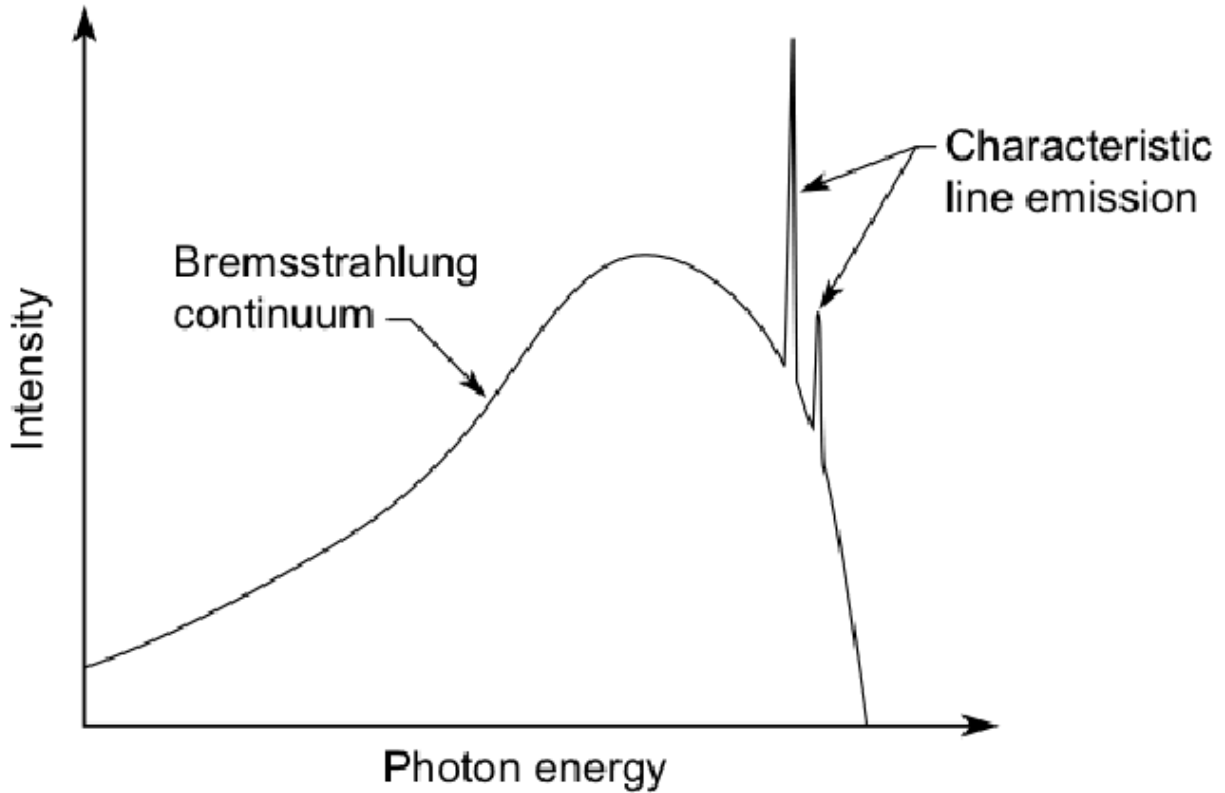


Fig 1. The power spectrum of the Bremsstrahlung radiation.

We can see that at high frequencies, the power is proportional to $e^{-h\nu/kT}$, which is called **the high frequency cutoff**, and this high frequency limit corresponds to optically **thin** plasma. At low frequencies, the gas is optically **thick**, and it becomes black body like, so the low frequency limit is simply the Rayleigh-Jeans limit, which is given by

$$I_\nu \sim \nu^2$$

Re-absorption at this limit is important. For the mid-frequency range, $I_\nu \text{ const}$ simgenerating a "flat platform range", and here we have

$$\epsilon_\nu \propto Z^2 n_e n_i T^{-1/2}$$

where Z is the charge number of the ion, ϵ_ν is the volume emissivity.

Besides, there also exists free-free absorption, but it is usually weaker than emission and the absorption coefficient has $\alpha \propto \nu^2$. If you look at temperature range higher than $10^6 - 10^7 \text{K}$, **the radiation is dominated by the Bremsstrahlung radiation**, and you can see the X-ray emission from this hot gas.

2. Thermal states of the ISM

Cooling

First of all, there is a thermal process called **cooling**. Cooling is a total description of the process which can decrease the temperature of the gas and "cool it down". With no loss of generality here we consider the optically thin gas.

Mathematically, cooling describes the net energy loss per unit time per unit volume, which is given by $\mathcal{L} = n^2 \Lambda(T)$, where $\Lambda(T)$ is the cooling function, or usually referred as **cooling curve**, which is in unit of $\text{erg cm}^3 \text{s}^{-1}$. The n^2 here means the cooling is simply a two-body process.

We can define a cooling timescale by $t_{\text{cool}} = \frac{E_{\text{th}}}{\mathcal{L}} = \frac{kT}{n\Lambda(T)}$. One cooling curve is given as follows

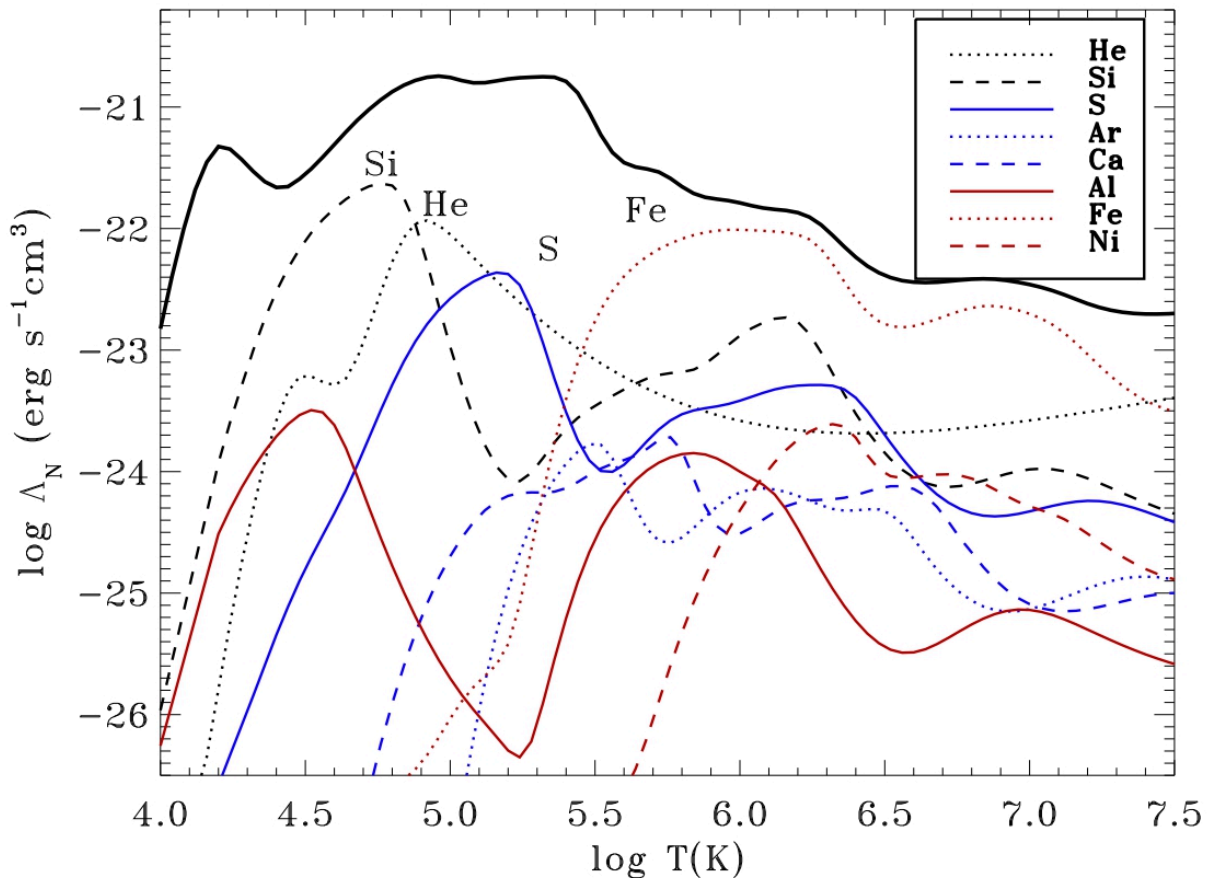


Fig 2. The cooling curve from Schure et al. (2009)

From the above figure, we can have some discussion on the cooling curve:

For $T \geq 10^7 K$, all elements are fully ionized, so **the cooling is dominated by the Bremsstrahlung radiation**, which is given by $\Lambda(T) \propto T^{1/2}$.

For $T \approx 10^4 - 10^7 K$, there exists excitation of electron, and after the excitation the electrons can spontaneous de-excite and emit photon, which is called as **the forbidden line**. The forbidden line will only occur in the dilute ISM not the dense gas like the atmosphere of the earth, the reason is, in a dilute media the excited electrons can survive for a long time without colliding with ions and de-excite by this collision. So in most times they can spontaneously de-excite and emit photon. While in the dense ISM, the excited electrons will be collisionally de-excited before they can spontaneously de-excite, in this case cooling is not efficient.

The excitation process is mainly dominated by **metals**, so it is also called **the metal line cooling**. If the gas is **metal rich**, the cooling is much more efficient than the metal poor media.

For $T \approx 10^4 K$, the main cooling process is the **hydrogen excitation**.

If we go to low temperatures of $T < 10^4 K$, the excitation of **fine structure** will dominate the cooling. Since when temperature is low, the particles do not have enough energy to let the electron get rid of the nucleus, but they can excite the electron into a slightly higher energy level called the "fine structure", and then the electron can drop back to its original level and emit a photon. The emission usually is in the band of infrared or sub-millimeter. One typical example is the molecular cloud we mentioned in the previous chapters.

If we go to high temperatures of $T > 10^9 K$, the cooling is dominated by **the pair production**. In extremely high temperature regime, the gas can be considered as in thermal equilibrium with the radiation field which is mainly constituted by high energy gamma rays. The gamma ray can collide with each other and generate an electron-positron pair, which will decrease the energy of the radiation field. And since the gas is tightly coupled with the radiation field, the gas will cool down to compensate for this loss of energy of the radiation field, this is the so-called **pair production cooling**. However, in real ISM nearly no gas can reach such a high temperature, except for the gas who is in a

special environment like in the accretion disk around a black hole, so in most cases this mechanism is not important for the ISM.

Heating

Secondly, there also exists a thermal process named heating. We know from previous chapters that the dominant energy in ISM involves the thermal energy, the turbulent energy, the cosmic ray energy, the magnetic energy, and the photon energy. These energies can be converted into each other, and the conversion from these energies to thermal energy is called **heating**.

First of all, there exists the **turbulent heating/dissipation**. The turbulent cascade lies in the heart of turbulence theory. This process means the energy is usually injected at large scale and after a range of inertial scale where energy conservation is satisfied, at some small scale named dissipation scale the turbulent kinetic energy will be converted into thermal energy by the viscosity of the gas.

Second, the **cosmic ray heating**. We will not discuss it in detail today and leave them to the future lectures.

Third, the **magnetic reconnection**. The magnetic reconnection can convert the magnetic energy into thermal energy, and this is very important for the heating of the ISM. The magnetic reconnection can occur near the surface of the star and heat the gas.

Finally, the **photoionization heating and photoelectric heating**. Usually we can find these photoionization heating around young stars and AGN.

- The photoionization heating simply speaking is the process that the UV photon can ionize hydrogen atom and transfer the excess energy of the bind energy into the kinetic energy of the electron, which will heat the gas.
- The photoelectric heating is the process that the UV photon can hit the dust grain and get absorbed. The dust grain will then emit a photoelectron, and the kinetic energy of this photoelectron will heat the gas.

We do not give a detailed description of heating like what we do for the cooling since heating is much more complicated and much more difficult to handle than cooling. To give

you a sense of this fact, we have some clues. First clue, the heating dependence on parameters is much more complicated, unlike the cooling which is usually dependent on n^2 as we mentioned before, heating can depend on n , n^2 , V and even cannot find such dependency. One important reason for this is there are many different heating sources and they usually have different heating mechanisms. Second clue, the heating and cooling are not balanced in general so we cannot derive the heating rate directly from the cooling rate.

HII regions (spherical chicken)

The HII regions are the ionized regions around the young (O/B) stars. We can somehow derive the radius of the HII region analytically, but in an extremely simplified "Spherical Chicken" model. The average of the intensity over the solid angle Ω is given by

$$J(\nu, r) \approx \frac{L_\nu}{4\pi r^2} e^{-\tau(\nu, r)}$$

With that, let's consider what kind of photons can ionize the ISM. The answer is straightforward, as we all know, photons have energy higher than 13.6 eV can ionize the hydrogen. And the cross section of photon-hydrogen collision is given by

$$\sigma_{\text{ion}}(\nu) \approx 1.6 \times 10^{-18} \left(\frac{\nu}{\nu_1} \right)$$

where ν_1 is the frequency corresponding to 13.6 eV. Then the photoionization rate per atom is given by

$$\phi = \int_{\nu_1}^{\infty} \frac{4\pi J(\nu)}{h\nu} \sigma_{\text{ion}}(\nu) d\nu$$

The recombination rate to level i is given by

$$n_e \alpha_i(T) = \int v \sigma_{\text{rec}, i} f(v) dv$$

We require the photoionization equilibrium (PIE), which is given by

$$n_e n_{\text{HII}} \alpha(T) = n_{\text{HI}} \phi$$

More generally, we have

$$R_i = \frac{\partial n_i}{\partial t} = n_i(n_{i+1}\alpha_{i+1} + n_{i-1}C_{i-1} - n_i(\alpha_i + C_i)) + (n_{i-1}\phi_{i-1} - n_i\phi_i)$$

Then we define the ionization state by

$$x = \frac{n_{\text{HII}}}{n}$$

where n is the total number density of hydrogen. Then finally we can get the equation for the ionization state, which is given by

$$\frac{1-x}{x^2} = \frac{n\alpha(T)}{\phi}$$

and we need to solve this equation. Before we really solve this equation, we can first observe the structure of this equation.

For the region close to the star, the photoionization rate is very high, so

$$\frac{n\alpha}{\phi} \ll 1$$

thus we have $1-x \ll 1$.

For the region far from the star, the photoionization rate is very low, so

$$\frac{n\alpha}{\phi} \gg 1$$

thus we have $x \ll 1$.

So the question is, where does the neutral transition occur? To answer this question more concisely, we need to bring back the concept of optical depth, which is given by

$$\tau(\nu, r) = \int_0^r n(1-x)\sigma_{\text{ion}}(\nu)dr' \approx n(1-x)\sigma_{\text{ion}}(\nu)r$$

we assume $n \sim \text{const}$ and $1-x \sim 0 \sim \text{const}$ here. The local intensity is given by

$$u(\nu, r) = \frac{J(\nu, r)}{c}$$

then we substitute $J(\nu, r)$ and $\tau(\nu, r)$ into this equation

$$J(\nu, r) \approx \frac{L_\nu}{4\pi r^2} e^{-\tau(\nu, r)}$$

and get

$$u(\nu, r) \approx \frac{L_\nu}{4\pi r^2 c} e^{-n(1-x)\sigma_{\text{ion}}(\nu)r}$$

We can get the equation

$$\frac{dS(r)}{dr} = -4\pi r^2 \alpha(T) n^2 x^2$$

where

$$S(r) = \int_{\nu_1}^{\infty} \frac{L_\nu}{h\nu} e^{-n(1-x)\sigma_{\text{ion}}(\nu)r} d\nu$$

is the number of ionizing photons per unit time that can reach the radius r .

If we assume the density of ionized gas is constant, we can derive the Strömgren radius, which is given by

$$R_S = \left(\frac{3S(0)}{4\pi\alpha(T)n^2} \right)^{1/3}$$

The Strömgren radius is the radius at which $S \rightarrow 0$. This means **the Strömgren radius is the radius where the ionization and recombination are balanced.**

From the formula we can see that there is no dependence on the cross section and the radius, it only depends on the total number of ionized photons, not their energy or cross section.